

Network Incident Analysis: TCP SYN Flood

Incident Summary

The network interruption appears to have been caused by a DoS attack, specifically a SYN flood. The packet capture shows a high number of SYN requests coming from the unfamiliar source IP address 203.0.113.0 and being sent to the web server at 192.0.2.1.

The server responds with SYN-ACK packets, but the connections are not being completed normally. This causes incomplete TCP connections to build up and consume server resources. As the server becomes overloaded, it becomes unable to reliably establish connections for legitimate users, resulting in connection timeout errors.

Technical Analysis and Impact

TCP connections normally use a three-way handshake. A client sends a SYN request, the server responds with SYN-ACK, and the client sends a final ACK to complete the connection.

In this case, the server is receiving a large number of SYN requests without the connections being properly completed. This leaves multiple connections in a half-open state and consumes resources that would normally be available for legitimate traffic.

The packet log shows repeated SYN requests from 203.0.113.0, along with SYN-ACK responses from the server. As these requests build up, fewer server resources remain available to process normal connection requests. This results in slow response times and connection timeouts for employees and customers trying to access the website.

If the attack continues, it could cause extended service outages and prevent users from accessing the company's sales page.